

DSA0503-Query Processing for Data Science

CO1 - Assessment Tool 2-Application based questions

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1. Online Examination System (CSV + XML)

Tasks:

1. Read student details from the **CSV** file and submission records from the **XML** file.
2. Compare both datasets to identify students who have **not submitted** their answers.
3. Generate a report displaying the list of non-submitted students along with the total submission statistics.

Code:

```
#Online Examination System
# Import required libraries
import pandas as pd
import xml.etree.ElementTree as ET

# Read student details from CSV
students = pd.read_csv("students.csv")

# Read submission details from XML
tree = ET.parse("submissions.xml")
root = tree.getroot()

submitted_students = []

# Extract submitted student IDs
for submission in root.findall("Submission"):
    student_id = submission.find("StudentID").text
    status = submission.find("Status").text
```

```

        if status == "Submitted":
            submitted_students.append(student_id)

# Find students who have not submitted
not_submitted =
students[~students["StudentID"].isin(submitted_students)]

# Display report
print("==== ONLINE EXAMINATION REPORT =====\n")

print("Students who have NOT submitted:\n")
print(not_submitted.to_string(index=False))

print("\nTotal Students :", len(students))
print("Submitted      :", len(submitted_students))
print("Not Submitted   :", len(not_submitted))

```

Output:

```

==== ONLINE EXAMINATION REPORT =====

Students who have NOT submitted:

StudentID  Name  Department
      S102 Priya    AI&DS
      S104 Meena      IT
      S106 Divya    AI&DS

Total Students : 7
Submitted      : 4
Not Submitted   : 3

```

2. Railway Ticket Reservation (CSV + JSON)

Tasks:

1. Read passenger details from the **CSV** file and train schedule details from the **JSON** file.
2. Verify whether each passenger's train number exists in the train schedule.
3. Generate a reservation report showing the train name and reservation status (**Valid/Invalid**) for each passenger.

Code:

```
#Railway Reservation
# Import required libraries
import pandas as pd
import json

# Read passenger details
passengers = pd.read_csv("passengers.csv")

# Read train schedule
with open("trains.json", "r") as file:
    train_data = json.load(file)

# Create dictionary of train numbers
train_dict = {}

for train in train_data["trains"]:
    train_dict[str(train["TrainNo"])] = train["TrainName"]

print("==== PASSENGER RESERVATION REPORT =====\n")

# Verify train numbers
for index, row in passengers.iterrows():

    train_no = str(row["TrainNo"])

    if train_no in train_dict:
        train_name = train_dict[train_no]
        status = "Valid"
    else:
        train_name = "Not Found"
```

```
status = "Invalid"

print(f"{row['PassengerID']} {row['Name']} {train_no} {train_name} {status}")
```

Output:

```
===== PASSENGER RESERVATION REPORT =====

P101  Ravi  12627  Karnataka Express  Valid
P102  Sneha 12631  Nellai Express   Valid
P103  Kumar 12627  Karnataka Express  Valid
P104  Divya 12345  Not Found    Invalid
P105  Arjun 12622  Tamil Nadu Express Valid
P106  Meena 12631  Nellai Express  Valid
P107  Karthik 12650  Not Found    Invalid
```